# Conduction, convection, radiation

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Identify mechanisms of heat transfer and support each classification.

**SUCCESS CRITERIA - I can...**

1. Describe contact transfer, bulk fluid motion and electromagnetic transfer.
2. Use specific details or measurements from the situation.
3. Recognize that more than one mechanism can act.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “conduction” and “convection” to explain your answer. Sentence starters are optional.

**conduction**: energy transfer through microscopic interactions in matter

**convection**: energy transfer through bulk motion of a fluid

**radiation**: energy transfer by electromagnetic waves

## 01 Warm-up: bring back an earlier idea

A device transfers **15 J** in **3 s**, then performs the same transfer in **6 s**. **Find both average powers** and **explain the difference**.

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $E_i + E_{in} = E_f + E_{out}$.
Account for energy across the system boundary. Conduction: matter interactions; convection: fluid motion; radiation: electromagnetic waves.

**Worked reasoning:** A metal rod transfers energy from its hot end toward its cool end by conduction. A circulating fluid can carry energy by convection; radiation can cross a vacuum.

**Your turn:** 1. **Draw a separate**, labeled **diagram** for each way energy is transferred.
2. **Show the direction of each** energy transfer.

**Sentence starter:** In my work, I used ... to show ... . I labeled it ... .

| Conduction | Convection | Radiation |
|---|---|---|
| Sketch: | Sketch: | Sketch: |
| _____ | _____ | _____ |

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

Classify: a metal spoon warming in soup; circulating water in a pot; sunlight warming your hand. **Give a reason** for each.

**Sentence starter:** I notice ... . This supports ... because ... .

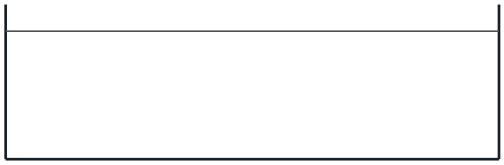
| Situation | Mechanism | Evidence |
|---|---|---|
| Spoon in soup | ____ | ____ |
| Circulating water | ____ | ____ |
| Sunlight on skin | ____ | ____ |

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A pot of water is heated from below. On the diagram, draw a circulation loop: warmer, less-dense water rising and cooler water sinking. **Explain how this bulk motion** differs from energy transfer along a solid metal spoon.

**Sentence starter:** Both ... . They differ in ... because ... .



Heater below pot

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Explain your reasoning

An asteroid travels through nearly empty space. Which of these mechanisms can transfer thermal energy to it from sunlight? Why not ordinary air convection?

**Sentence starter:** I think … . I can point to …, which shows …, so … .

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A rolling object loses **24 J** gravitational energy but gains only **16 J** translational kinetic energy. **Name two possible destinations of** the difference.
**Show the energy changes** and transfers; **label units**.

**Sentence starter:** Both … . They differ in … because … .

**Initial** | **Transfers** | **Final**

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3

## 07 Exit check

**Explain how energy can travel** from the Sun to Earth through space.

**Sentence starter:** I think … . I can point to …, which shows …, so … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

1. Design insulation for a hot cup using one change for each transfer mechanism.
2. **Explain each change**.

**Sentence starter:** My design includes … . It does what is needed by … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** 1 2 3